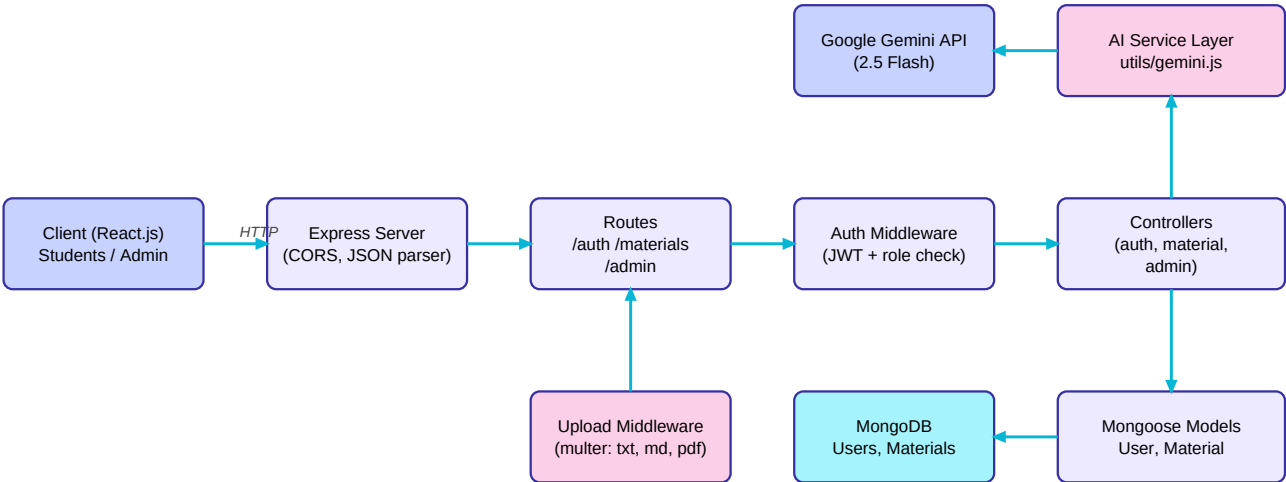


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

Technical Architecture:



Technical architecture of AI StudyBuddy

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web client where students and admins interact with the application	React.js (frontend); Postman / Thunder Client for API testing
2	Application Logic-1	Server, routing, controllers and middleware	Node.js, Express.js
3	Application Logic-2	Authentication and authorization	JWT (jsonwebtoken), bcryptjs, role check middleware
4	Application Logic-3	Study material upload handling	multer (.txt, .md, .pdf, max 5 MB)
5	Database	Users and study materials with embedded AI resources	MongoDB with Mongoose ODM
6	Cloud Database	Managed database hosting	MongoDB Atlas
7	File Storage	Uploaded files saved to the uploads folder on the server	Local filesystem (uploads/)
8	External API	Generates educational content from study material	Google Gemini API (2.5 Flash) via @google/genai

S.No	Component	Description	Technology
9	Machine Learning Model	Large language model for summaries, flashcards, quizzes and plans	Google Gemini 2.5 Flash
10	Infrastructure (Server / Cloud)	Local: Node.js v16+, npm v8+, Windows/macOS/Linux, 8 GB RAM (16 GB recommended), 1 GB disk. Cloud: any Node.js host with MongoDB Atlas	Local server (PORT 5000); cloud deployment optional

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source runtime, framework, ODM and libraries	Node.js, Express.js, Mongoose, cors, dotenv, multer, nodemon
2	Security Implementations	Stateless auth, password hashing, role-based access, upload validation, secrets in environment variables	JWT, bcryptjs, RBAC, .env, CORS
3	Scalable Architecture	Modular MVC layers with an isolated AI service	Routes / controllers / models / middleware / utils
4	Availability	Stateless server instances behind a load balancer with a managed database	Load balancer (deployment option), MongoDB Atlas
5	Performance	Asynchronous AI calls, upload size limit and lightweight routing	Express, async/await, multer limits